

AVINAV KUMAR

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EDUCATION

SRM Institute of Science and Technology
B.Tech in Computer Science and Engineering
· CGPA: **9.91**

2024 – 2028
Kattankulathur, Tamil Nadu

DAV Hehal
Class XII
· Percentage: **87.16%**

2020 – 2022
Ranchi, Jharkhand

WORK EXPERIENCE

Green Agrevolution Pvt. Ltd.
SDE Intern

June 2025 – July 2026

- Developed a custom ERP application for nursery operations using the Frappe Framework and ERPNext.
- Designed custom DocTypes, workflows, dashboards, and business forms to digitize nursery management.
- Integrated WhatsApp automation, REST APIs, Python, JavaScript, and MariaDB for business process automation.

Apollo Tyres R&D Global Asia – Phase III Project

April 2026 – Present

- Machine Learning Industry Research Collaboration
- Processed over **500M+ GPS and TPMS** records to extract **10,000+ tyre duty cycle events** for fleet-scale thermodynamic analysis.
- Built an end-to-end data engineering pipeline synchronizing GPS and TPMS data, reducing manual engineering effort by nearly **90%**.
- Proposed machine learning models for pressure build-up prediction, tyre family comparison, load sensitivity analysis, and failure prediction.

PROJECTS

Warehouse Occupancy Estimation Platform

- Built a production-ready warehouse occupancy estimation platform using FastAPI, Streamlit, PostgreSQL, SQLAlchemy and APScheduler.
- Integrated pretrained SegFormer semantic segmentation with ROI-based processing and RTSP multi-camera architecture.
- Developed scalable REST APIs, weighted occupancy aggregation, structured logging, Alembic migrations and camera health monitoring.

WanderLust

- Developed a full-stack travel and accommodation platform using Node.js, Express.js, MongoDB, EJS and Bootstrap.
- Implemented CRUD operations, authentication using Passport.js, image uploads with Multer and flash messaging.
- Deployed the application on Render with secure environment variable management.

Mann Ki Baat

- Engineered a full-stack WebRTC video platform using React and Node.js, achieving sub-200ms signaling latency.
- Architected a decentralized WebRTC Mesh network supporting seamless peer-to-peer streaming for concurrent users.
- Designed a responsive CSS Grid UI handling 50+ real-time WebSocket events per second at a stable 60 FPS.

TECHNICAL SKILLS

Languages	C, Java, Python, JavaScript, HTML, CSS
Frameworks	React.js, Node.js, Express.js, Tailwind CSS, Bootstrap, Frappe, ERPNext
Databases	MongoDB, MySQL, PostgreSQL, Supabase, MariaDB
Libraries	NumPy, Pandas, Matplotlib
Tools	Git, GitHub, Docker, REST APIs
Coursework	Data Structures, Algorithms, DBMS, Operating Systems, OOP
Areas of Interest	AI/ML, Full Stack Development, Data Engineering, Problem Solving

ADDITIONAL ACHIEVEMENTS

- Best Team – HackPick Hackathon, SRMIST (August 2025)
- Winner – Samsung PRISM Metaverse AR/VR Ideathon (2025–2026)
- Silver Elite – NPTEL Fundamentals of Object-Oriented Programming